Maize (corn) - Common rust

RAG record ADV-001 | canonical key: maize_common_rust

Record ID	ADV-001
Crop / crop key	Maize (corn) / maize
Condition / disease key	Common rust / common_rust
Main pathogen	Puccinia sorghi
RAG aliases	maize common rust, corn common rust, Puccinia sorghi, rust pustules

Farmer quick guide: treatments protect healthy plant tissue and slow disease spread. Dead or badly damaged leaf tissue will not become green again.

How to recognize it

- Look for many small, raised, cinnamon-brown powdery pustules on both sides of the leaf.
- Older pustules can turn dark brown or almost black. Heavy infection reduces the green leaf area.

Weather or field conditions that favour it

Cool to mild, moist weather (about 15-25 C) and long periods of leaf wetness favour spread.

What to do now

- Check several plants across the field. Rust pustules should rub off as a brown powder.
- Remove badly infected crop residue after harvest and use resistant/tolerant hybrids where available.
- If disease is increasing on a susceptible crop, use a currently registered fungicide rather than waiting until most leaves are affected.

Chemical treatment - only after the disease is confirmed

Chemical / formulation	Rate from cited source	How to use this information	Source
Kresoxim-methyl 44.3% SC	1 ml/L water	Foliar option listed by TNAU for maize common rust; TNAU places sprays around 35	[S1]

Chemical / formulation	Rate from cited source	How to use this information	Source
		and 50 days after sowing in its guidance.	
Tebuconazole	1 ml/L water	Alternative foliar option listed by TNAU. Use only a product whose current label allows maize/rust.	[S1]

Safety and compliance: Use only a product whose current Indian label allows the crop and target disease. Follow the label rate, water volume, re-entry interval and pre-harvest interval. Wear label-required PPE. Do not mix products unless the label or a qualified extension adviser permits it.

Prevention and integrated management

- Use resistant hybrids where locally recommended.
- Destroy or bury infected residues after harvest.
- Avoid unnecessary fungicide sprays when disease pressure is low.

RAG guardrail

The AI should treat the chemical table as source-grounded examples, not permission to invent a different pesticide or dose. If the retrieved source does not match the crop, disease and formulation, return non-chemical advice and request local verification.

Sources used for this record

[S1] Tamil Nadu Agricultural University (TNAU). Maize - Common Rust: Puccinia sorghi.
https://www.agritech.tnau.ac.in/crop_protection/maize_disease_new/maize_4.html